FOOD DELIVERY DATA ANALYSIS

Business Insights, Operational Findings & Recommendations

SQL Analysis Report | 45,593 Records | February–April 2022

Prepared from 20 core SQL analyses and Stage 1 data validation

1. Project Overview

This project analyzes 45,593 food delivery records using MySQL. Its objective is to understand delivery performance, identify operational challenges, investigate factors associated with delivery delays, and develop data-driven recommendations.

The analysis covers traffic, weather, vehicle conditions, order types, delivery-person characteristics, festivals, city categories, order timing, and delivery workload.

2. Key Business Findings

2.1 Traffic Congestion and Delivery Performance

Traffic condition	Orders	Average delivery time
Low	15,477	21.27 min
Medium	11,548	26.69 min
High	4,425	27.24 min
Jam	14,143	31.18 min

Deliveries under Jam conditions take approximately 9.91 minutes longer on average than deliveries under Low traffic conditions.

Business implication: Consider traffic-aware route planning, congestion hotspot monitoring, and more realistic delivery-time estimates during congested periods.

2.2 Weather Conditions and Delivery Time

Weather	Orders	Average delivery time
Cloudy	7,536	28.92 min
Fog	7,654	28.92 min
Windy	7,422	26.12 min
Sandstorms	7,495	25.88 min
Stormy	7,586	25.87 min
Sunny	7,900	22.22 min

Cloudy and Fog conditions have the highest average delivery times; Sunny conditions have the lowest.

Business implication: Consider weather-sensitive delivery estimates and operational planning.

2.3 Multiple Deliveries and Delivery Delays

Multiple deliveries	Orders	Average delivery time
0	15,088	22.87 min
1	28,159	26.86 min
2	1,985	40.45 min
3	361	47.82 min

Average delivery time rises across the multiple-delivery groups. The difference between 0 and 3 is approximately 24.95 minutes.

Business implication: Review order batching, assignment limits, and the geographical compatibility of simultaneously assigned orders.

2.4 Festival Periods and Delivery Performance

Festival status	Orders	Average delivery time
Yes	896	45.52 min
No	44,697	25.91 min

Festival-period deliveries average 19.61 minutes longer.

Business implication: Consider additional delivery capacity, workforce planning, and customer communication during festivals.

2.5 Hourly Order Demand

- Highest order volume: 9 PM, with 4,686 orders.
- Longest average delivery time: 8 PM, at 31.19 minutes.
- Shortest average delivery time: 10 AM, at 19.48 minutes.

Business implication: Review delivery-person availability during evening peak hours, particularly 7 PM–9 PM.

2.6 City-Wise Delivery Performance

City category	Orders	Average delivery time
Semi-Urban	164	49.73 min
Metropolitian	34,093	27.32 min
Urban	11,336	22.89 min

Semi-Urban has the highest average delivery time, but represents only 164 records; interpret this result cautiously.

Business implication: Investigate delivery distances, restaurant coverage, and delivery-person availability before making location-specific changes.

2.7 Vehicle Type and Delivery Performance

Vehicle type	Orders	Average delivery time
Motorcycle	26,435	27.61 min
Bicycle	68	26.43 min
Scooter	15,276	24.48 min
Electric scooter	3,814	24.47 min

Motorcycles have the highest average among vehicle categories with substantial representation. Bicycle results are based on only 68 records.

Business implication: Evaluate vehicle allocation alongside traffic, delivery distance, and route conditions rather than relying on vehicle type alone.

2.8 Vehicle Condition and Delivery Performance

Vehicle-condition code 0 averages 30.07 minutes, compared with 24.36 minutes for code 1 and 24.46 minutes for code 2.

Business implication: Investigate vehicle maintenance and condition. The meanings of the numeric codes have not been verified, so no descriptive labels are assigned.

2.9 Order Type and Delivery Performance

Order type	Orders	Average delivery time
Meal	11,458	26.42 min
Snack	11,533	26.29 min
Buffet	11,280	26.28 min
Drinks	11,322	26.19 min

The difference between the highest and lowest averages is only 0.23 minutes.

Business implication: Order type shows relatively little difference in average delivery time; prioritize traffic, weather, and delivery allocation.

3. Additional Findings

- Average delivery-person age: 29.58 years.
- Average delivery-person rating: 4.62.
- Wednesday had the highest order volume and longest average delivery time among weekdays; Thursday had the shortest average delivery time.
- The highest recorded workload among the top 20 delivery persons was 67 orders.
- Rating groups showed differences in average delivery time, but some groups had very small samples.

4. Business Recommendations

- **Traffic-aware planning:** Use traffic information to identify congestion hotspots, improve route selection, and adjust delivery estimates.
- **Weather-sensitive estimates:** Incorporate weather conditions into delivery-time forecasting and operational planning.
- **Review order batching:** Evaluate whether multiple simultaneous deliveries create excessive delays; review assignment limits and route compatibility.
- **Festival readiness:** Plan staffing and delivery capacity ahead of festival periods.
- **Evening workforce allocation:** Review delivery-person availability around 7 PM–9 PM.
- **Semi-Urban investigation:** Examine distances, restaurant coverage, and workforce distribution before targeted interventions.
- **Vehicle maintenance:** Investigate vehicle-condition differences and establish regular maintenance monitoring.
- **Prioritize operational factors:** Focus on traffic, weather, batching, and capacity planning rather than order type.

5. Data Quality and Validation

Validation finding	Result
Total records	45,593
Missing values in nine checked columns	0
Duplicate order IDs	0
Invalid order dates	0
Invalid order times	0
Invalid non-positive delivery times	0
Ratings outside expected 1–5 range	53
Delivery-person age of 15	38 records
Delivery times above 45 minutes	1,562 records

Treatment of suspicious values:

- The 53 ratings of 6.0 are flagged as suspicious because they exceed the expected 1–5 scale. Original values are preserved; no automatic replacement with 5.0 is made.
- The 38 records with delivery-person age 15 are flagged for review and have not been automatically deleted or changed.
- Delivery times above 45 minutes are retained because a long delivery time is not automatically an invalid record.
- The original dataset has not been modified.

6. Analytical Limitations

- The analysis identifies associations, not proven causal relationships.
- Semi-Urban and Bicycle categories have relatively small samples.
- The meanings of vehicle-condition codes have not been verified.
- Delivery distance and other potentially influential factors were not fully controlled for.
- Suspicious ratings and age values require further investigation.
- The dataset covers February 11 to April 6, 2022; findings may not represent current operating conditions.

7. Executive Summary

The analysis of 45,593 food delivery records indicates that traffic congestion, weather conditions, multiple deliveries, festival periods, and evening demand are associated with delivery performance. Average delivery time increases from 21.27 minutes under Low traffic conditions to 31.18 minutes under Jam conditions. Festival-period deliveries average 45.52 minutes compared with 25.91 minutes outside festivals. Multiple deliveries also show a substantial increase in average delivery time.

The findings suggest that traffic-aware routing, reviewing delivery batching, festival capacity planning, and evening workforce allocation may help improve operations. Validation found no missing values in the nine checked columns and no duplicate order IDs, but identified 53 suspicious ratings of 6.0 and 38 records with delivery-person age 15. These records have been preserved for review. The project demonstrates how SQL analysis can produce operational insights while acknowledging data-quality limitations.

8. Conclusion

This project demonstrates SQL-based data exploration, aggregation, validation, and business analysis. Across 20 analytical questions and a structured validation process, it identifies operational patterns and develops practical recommendations for improving food delivery performance.

The findings provide a foundation for further investigation, dashboard development, and data-driven decision-making.